

# The Standard Error

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## Introduction

The **standard error** of a statistic is the standard deviation of its sampling distribution. It quantifies how much the statistic would vary across hypothetical repetitions of the sampling process. Every confidence interval and every t- or z-statistic is built on an estimate of the standard error, so understanding and correctly computing SEs is central to competent data analysis.

## Prerequisites

A reader should know what the sampling distribution is, what a sample standard deviation is, and how to use the `sd()` and `mean()` functions in R.

## Theory

For an iid sample  $X_1, \dots, X_n$  from a population with variance  $\sigma^2$ , the sample mean  $\bar{X}$  has variance  $\sigma^2/n$  and therefore

$$\text{SE}(\bar{X}) = \sigma/\sqrt{n}.$$

Because  $\sigma$  is usually unknown, it is estimated by the sample SD  $s$ , giving the plug-in estimator  $\widehat{\text{SE}}(\bar{X}) = s/\sqrt{n}$ .

For a binary proportion  $\hat{p} = k/n$ , the standard error under binomial sampling is

$$\text{SE}(\hat{p}) = \sqrt{p(1-p)/n}, \quad \widehat{\text{SE}}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}.$$

For a difference in means across two independent samples with variances  $s_1^2, s_2^2$ ,

$$\widehat{\text{SE}}(\bar{X}_1 - \bar{X}_2) = \sqrt{s_1^2/n_1 + s_2^2/n_2}.$$

For regression coefficients, standard errors come from the diagonal of  $\hat{\sigma}^2(X^\top X)^{-1}$ ; R returns them in `summary(fit)$coefficients`.

For statistics without a closed-form SE (ratios, quantiles, custom summaries), the **bootstrap** gives a general-purpose estimator: resample with replacement  $B$  times, compute the statistic for each resample, and use the SD of the  $B$  bootstrap statistics as  $\widehat{\text{SE}}$ .

## Assumptions

The formula-based SEs above assume iid sampling and, for the finite-sample versions, enough regularity for the CLT to justify using them in normal-approximation CIs. The bootstrap SE is consistent under weaker conditions but inherits the design: it is accurate only when resampling units are the same as the independent sampling units (so bootstrap at the cluster level for cluster samples).

## R Implementation

```
library(boot)

set.seed(2026)
x <- rnorm(60, mean = 75, sd = 12)

se_mean <- sd(x) / sqrt(length(x))
se_mean

y <- c(rep(1, 24), rep(0, 76))
p_hat <- mean(y)
se_prop <- sqrt(p_hat * (1 - p_hat) / length(y))
se_prop

iqr_stat <- function(d, i) IQR(d[i])
b <- boot(x, statistic = iqr_stat, R = 5000)
se_iqr <- sd(b$t)
se_iqr
```

The first two blocks use the closed-form SEs for the mean and a proportion; the third uses the bootstrap for the IQR, which has no convenient closed form.

## Output & Results

```
[1] 1.552      # SE(mean)
[1] 0.0427     # SE(proportion)
[1] 2.11      # bootstrap SE(IQR)
```

Each of these is the estimated typical distance between the sample statistic and the population parameter it estimates.

## Interpretation

The standard error is what gets reported alongside the estimate: “mean 75.8 (SE 1.6)” or “proportion 0.24 (95% CI 0.16–0.32)”, where the CI is approximately the estimate plus/minus 1.96 times the SE. A reviewer glancing at the SE can judge whether the study is precise enough to support its claims.

## Practical Tips

- Always distinguish the SD (spread of individual values) from the SE (spread of a statistic); papers regularly report the wrong one.
- For proportions near 0 or 1, the normal-approximation SE is poor; prefer Wilson or Clopper-Pearson intervals, which are available via `binom.test()` and the `binom` package.
- For differences between groups, the correct SE is the SE of the difference, not either group’s SE by itself.
- When you lack a closed-form SE, reach for the bootstrap rather than inventing one; `boot::boot()` handles the resampling machinery robustly.
- Do not report an SE without stating what statistic it refers to – “SE = 1.6” alone is ambiguous; “SE of the mean = 1.6” is clear.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.

- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr